





Max Product Sub Array

WHAT IS SUB ARRAY

A subarray, also known as a contiguous subarray, is a sequence of elements within an array that preserves the order of the elements. In other words, a subarray is formed by selecting a range of elements from the original array without changing their relative order.

For example, consider the following array:

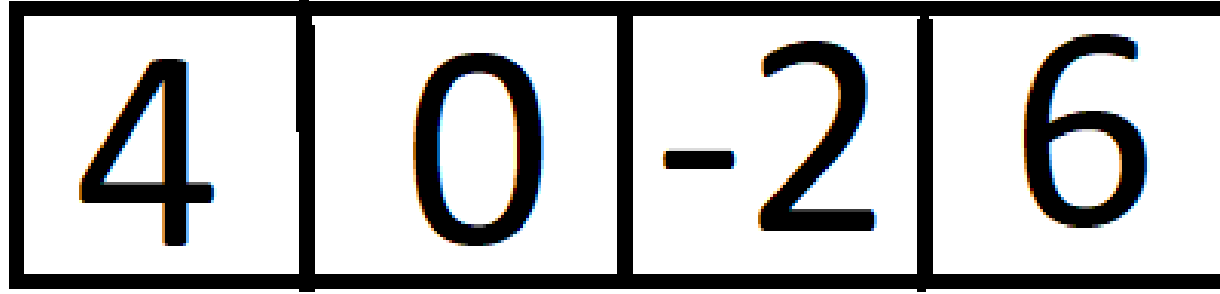
[1, 2, 3, 4, 5]

Possible subarrays of this array include:

- [1]
- [1, 2]
- [2]
- [1, 2, 3]
- [2, 3]
- [1, 2, 3, 4]
- [3]
- [2, 3, 4]
- [1, 2, 3, 4, 5]
- [3, 4]
- [2, 3, 4, 5]
- [4]
- [3, 4, 5]
- [4, 5]
- [5]

And so on. Each subarray maintains the order of the elements as they appear in the original array.

Subarrays are commonly used in algorithms and programming when dealing with array manipulation or searching for specific patterns or sums within an array.



The subarrays for the array {4, 0, -2, 6} are:

[4]			
	[4, 0]		
[0]		[4, 0, -2]	
	[0, -2]		[4, 0, -2, 6]
[-2]		[0, -2, 6]	
	[-2, 6]		
[6]			

In general, for an array of length N , there are $N(N+1)/2$ subarrays.

The subarrays for the array {4, 0, -2, 6} products are:

$$[4] \times 1 = 1$$

$$[4 \times 0] = 0$$

$$[0] \times 1 = 0$$

$$[4 \times 0 \times -2] = 0$$

$$[0 \times -2] = 0$$

$$[4 \times 0 \times -2 \times 6] = 0$$

$$[-2] \times 1 = -2$$

$$[0 \times -2 \times 6] = 0$$

$$[-2 \times 6] = -12$$

$$[6] \times 1 = 6$$

The max product of sub array is 6

```
1. import java.util.*;
2. public class Maxproduct
3. {
4. public static void main (String[] args)
5. {   int a[]={-5,-4,-1,-2};
6.     int max=a[0],m=a[0],min=a[0],temp=0;
7.     for(int i=1;i<4;i++)
8.     {
9.         if(a[i]>0)
10.        {
11.            max=Math.max(a[i],max*a[i]);
12.            min=Math.min(a[i],min*a[i]);
13.        }
14.        else if(a[i]==0)
15.            max=min=0;
16.        else
17.        {
18.            temp=max;
19.            max=Math.max(a[i],min*a[i]);
20.            min=Math.min(a[i],temp*a[i]);
21.        }
22.        m=Math.max(m,max);
23.    }
24.    System.out.println(m);}}
```



THANK YOU